Soma Sree Ved

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.08
Resonance Junior College	Class XII (TSBIE)	2024	95.70%
Sancta Maria International School, Hyderabad	Class X (IGCSE)	2022	90.14%

POSITIONS OF RESPONSIBILITY

Events Head ,Milan 2026 -Directing end-to-end execution of Milan, IIT Hyderabad's general championship spanning sports, cultural and technical events -- leading the 18-membered events team , managing logistics, and making real-time crisis decisions during live operations drawing 6,000+ attendees.

Internship Coordinator, Office of Career Services -Building and managing the corporate internship pipeline - initiating company outreach, coordinating job descriptions, tests and interviews, tracking student applications, and bridging student queries with recruiter requirements.

RELEVANT COURSES

Core Materials Science & Metallurgy- Steelmaking, Phase Transformations, Mechanical Behaviour of Materials, Extractive Metallurgy, Materials Thermodynamics and Kinetics, Materials Synthesis and Characterization

Computational & Analytical Methods- Computational Methods in Materials Science, Imaging in Materials Science, Diffraction and Spectroscopy

Mathematics- Linear Algebra, Differential Calculus, Integral Calculus, Differential Equations

Applied & Interdisciplinary- Semiconductor Materials, Digital Fabrication, Artificial Intelligence, Physics of Solids, Materials Chemistry

PROJECTS

Project Hail Mary -- Live Algorithmic Trading System

- Backtested 2 years of data: 35.7% net annual return, 1.48 profit factor
- Executed 170 trades across 293 eligible trading days in backtesting
- Deployed fault-tolerant system with real capital via Zerodha Kite Connect API on DigitalOcean
- Built crash-recovery, lock-file concurrency control, and heartbeat monitoring for 24/7 uptime
- Engineered quad 4-step trailing stop-loss with real-time Telegram trade alerts
- Ported trailing-stop logic to C++ to explore low-latency execution tradeoffs

Project: Working Micro-Spectrometer Model

- Led separate student teams across sem 3 and sem 4 project cycles
- Built a 40x magnification optical microscope from scratch in sem 3
- Upgraded system into a fully functional microspectrometer with a separate team in sem 4
- Integrated endoscope camera with Python-based data processing for spectral analysis
- Captured high-resolution specimen images with simultaneous spectral characterisation

M-IP Slag Viscosity Simulation *(with Prof. Ashok Kamaraj, IIT Madras)*

- Built a Python model reproducing a published high-temperature slag viscosity method
- Validated flow behavior via governing flow equation across 7 ash samples, 4 incline angles
- Extended model with coupled heat-transfer and Arrhenius-based viscosity equations
- Simulated real furnace conditions: transient heating, melting, and Coulomb friction resistance
- Solved as a coupled differential system over a 40-minute per-sample furnace cycle

EXTRACURRICULAR

Calisthenics - training five days a week with focus on strength and mainly skills-based movements

Music - self-taught fingerstyle guitarist and singer with active performance experience

Portrait Sketching - practising detailed human face sketching as a creative pursuit

Athletics :

-- Silver (5000m, 1500m) at Milan 2024

-- Gold (5000m) + Silver (800m) at Diesta 2025

-- Bronze (5000m) at IITH 2nd Annual Half Marathon 2025.

SKILLS

Programming & Data Science - Python, NumPy, Pandas, Matplotlib, Scikit-Learn, OpenCV

Finance & Deployment - Algorithmic trading via Zerodha Kite Connect API, cloud deployment on DigitalOcean

Hardware & Design - 3D printing via Creality, CAD design via Solid Edge

Tools & Platforms - Git, GitHub, JIRA (Agile/Scrum), Jupyter Notebook